

Wai Lau

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Backend Services & Distributed Systems

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7 years of experience building and operating backend services at scale across commerce and finance: distributed systems, high-throughput financial platforms, and build-and-deploy infrastructure across GCP and AWS. Capable of owning complex services end-to-end with SLAs, SLOs and on-call shifts. An early adopter of LLM-assisted workflows, shipping safely for sensitive codebases and driving complex migrations.

Coinbase | Software Engineer · Feb 2022 – May 2026

Financial Platform

- Built and operated a high-throughput, low-latency financial ledger with Aeron, serving international and institutional clients.
- Built multi-currency, multi-platform financial flows managing risk, slippage, and compensation steps.
- Mapped relational database records into SBE encodings for the Aeron transport layer, working directly against the underlying schema and query behaviour.
- Drove a Cadence-to-Temporal migration, leveraging Buf to match gRPC APIs across services.
- Carried the payments on-call pager for transfer availability, throughput, and correctness; triaged and resolved production issues end to end.
- Used an LLM-assisted, multi-repo development workflow (Cursor with Sourcegraph and Buf MCP servers) to ship changes to financial systems safely and quickly, work that was otherwise slow and risky by hand.
- Mentored interns and served as onboarding buddy for new team members.

Build & Deploy

- Owned and improved the company-wide CI/CD pipeline; unified two distinct container build-and-deploy architectures into one system.
- Carried the build-and-deploy on-call pager, handling escalations where a broken pipeline blocked urgent deploys during active incidents.
- Reduced AWS costs by \$500K/year by graphing ECR container image dependencies and safely pruning unused storage.
- Designed and integrated a gRPC schema (proto) registry into the deployment pipeline.

Shopify | Production Engineer · Jun 2019 – Feb 2022

Search Platform

- Built active-active multi-region failover for Shopify-wide search on GCP, using Kubernetes and Google Cloud load balancing; scaled to ~150TB indexed, handling ~2K QPS steady and 30K at peak.
- Led a cross-functional effort to bring Search to the edge, cutting p99 latency from 400ms to 50ms for Australia and Europe.
- Carried the on-call pager for Shopify-wide search availability, responding to downtime and degradation in production.
- Parallelized and sharded the search test suite, halving total test runtime; built queuing infrastructure for high-volume indexing and bursty backfills.
- Worked with a Prometheus sidecar streaming service telemetry into Datadog for search monitoring and alerting.

Developer Infrastructure

- Expanded and maintained company-wide developer tooling.
- Built infrastructure for testing macOS and iOS software (Anka, Docker, nested VMs).

Nuance Communications | Internship · June 2017 – Feb 2018

- Built internal tooling and automation across speech recognition and NLP team
 - Custom Django cluster-management interfaces
 - Jenkins CI/CD pipelines deploying NLP models
 - Grafana dashboards for observability and monitoring

Projects

- **emet** — a privacy-sensitive, local-LLM memory map with an MCP frontend
- **exec-fn** — self-hosted planning agent with a nightly memory pipeline

Skills

Languages: Python, Go, Java, Ruby, JS

Backend & Data: Postgres, Kafka, Elasticsearch, Redis, Aeron, Snowflake, gRPC, Buf, Envoy

Infrastructure: GCP, AWS, Kubernetes, Docker, Terraform, Temporal, Buildkite

Observability: Grafana, Prometheus, Datadog, Splunk, Logstash

AI / LLM: Cursor, Claude Code, MCP (Sourcegraph, Buf), local LLM setups, LLM-assisted migration

Education

Concordia University | B.Eng., Software Engineering (Co-op). Dean's List.